WOW! Learning Club 1: Time, seasons and Carroll diagrams

Lesson 4-1
Book pages 92-95

💡 **Big idea:** The Earth **turns** every 24 hours (day and night) and goes **around the Sun** in 365 days (a year and the **seasons**). We learn the **months**, **seasons** and **telling the time**, and use a **Carroll diagram** to classify **types of energy**.

🎯 **I will learn:**
- Name the months and seasons
- Tell the time (o'clock, quarter past, half past, quarter to)
- Explain day, night and seasons
- Classify machines and places in a Carroll diagram

📖 Key words

season فصل (من السنة)	**spring** الربيع	**summer** الصيف	**autumn** الخريف
winter الشتاء	**quarter past** والربع	**half past** والنصف	**quarter to** إلا ربع
energy طاقة	**Carroll diagram** مخطط كارول (جدول تصنيف)		

Amazing Earth

- The Earth turns around every 24 hours. It's day on the side closest to the Sun and night on the side furthest away
- The time is different around the world: when it's 7 p.m. in Muscat, it's 11 a.m. in New York
- It takes 365 days for the Earth to go around the Sun = one year
- In Europe: summer is June, July and August. In Australia it's the opposite: summer is December, January and February

Months and seasons

- January, February, March, April, May, June
- July, August, September, October, November, December
- Four seasons: spring, summer, autumn, winter

Telling the time

- 4:00 It's four o'clock.
- 4:15 It's quarter past four.
- 4:30 It's half past four.
- 4:45 It's quarter to five.

Carroll diagrams (science)

- All machines need energy. Some make light, sound, heat or movement
- A Carroll diagram shows yes / no information in four boxes
- Example: a theatre needs sound and light. An art gallery needs light but doesn't need sound
- A toaster makes heat. A food mixer makes movement



2 Which months are in winter in Australia?

Amazing Earth

Why do we have day and night?

The Earth turns around every 24 hours, so different parts of the planet can see the Sun at different times. It's daytime on the side that is closest to the Sun. It's night on the side that is furthest away. This is why the time is different around the world. When it's 7 p.m. in Muscat, it's 11 a.m. in New York! 8 a.m. in London is 7 p.m. in New Zealand!

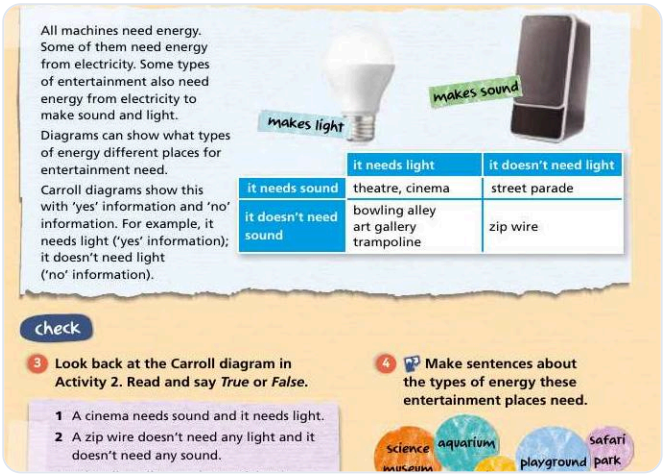
Why do we have seasons?

It takes 365 days for the Earth to go around the Sun. That's a year. It's summer in the countries that are closer to the Sun. It's winter in the countries that are further away. The calendar doesn't change around the world, but the seasons do. In Europe, there are four seasons: winter, spring, summer and autumn. June, July and August are in summer. December, January and February are in winter. Australia has the opposite seasons. So, when children are making snowmen in Europe, they're eating ice cream in Australia!

Did you know that time travel is possible?

It's true! You can experience the same day twice if you fly from New Zealand to Hawaii. For example, you can have 1st January in New Zealand. Then, if you get on a plane that takes off on 1st January, the plane will land in Hawaii on 31st December and you can experience 1st January again! You will need to pack a big suitcase. You won't need your flip-flops, but you will need your raincoat and umbrella! When it's summer in New Zealand, it's cold and rainy in Hawaii!

Language booster: Amazing Earth (p.92)



All machines need energy. Some of them need energy from electricity. Some types of entertainment also need energy from electricity to make sound and light.

Diagrams can show what types of energy different places for entertainment need.

Carroll diagrams show this with 'yes' information and 'no' information. For example, it needs light ('yes' information); it doesn't need light ('no' information).

	it needs light	it doesn't need light
it needs sound	theatre, cinema	street parade
it doesn't need sound	bowling alley art gallery trampoline	zip wire

check

3 Look back at the Carroll diagram in Activity 2. Read and say *True* or *False*.

1 A cinema needs sound and it needs light.
2 A zip wire doesn't need any light and it doesn't need any sound.

4 Make sentences about the types of energy these entertainment places need.

Think like a scientist: Carroll diagrams (p.94)

✏️ Examples

6:45

Answer: It's quarter to seven.

A street parade: light? sound?

Answer: It needs sound, but it doesn't need any light.

A cinema needs sound and light. True or false?

Answer: True

⭐ Remember

- Months always start with a **capital letter**: January, not january
- quarter **past** = after the hour, quarter **to** = before the next hour

⚠️ Watch out

- ✗ It's quarter past to five. ✓ It's quarter to five.
- ✗ In Australia summer is in July. ✓ It's in December, January and February.

🧩 **Activity:** Make a Carroll diagram of jobs: outside job / not outside job and work with animals / don't work with animals

✅ Check yourself

1. How long does the Earth take to turn around once?
2. How many days does it take to go around the Sun?
3. How do you say 9:30?
4. What does a Carroll diagram show?

Answers: (1) 24 hours ، (2) 365 days ، (3) It's half past nine ، (4) yes / no information to classify things